

hw2

1

①

Eigenval for AA^T : $AA^T = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$ $\begin{vmatrix} 2-\lambda & 2 \\ 2 & 2-\lambda \end{vmatrix} = (2-\lambda)(2-\lambda) - 4 = 4 - 4\lambda + \lambda^2 - 4 = \lambda^2 - 4\lambda = \lambda(\lambda-4)$ $\lambda_1=4, \lambda_2=0$

For V_1 , get eigvec of $A^T A$: $A^T A = AA^T$ b/c $A^T = A$ so $V_1: \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} V_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ normalize

$V_2: \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} \text{ or } \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} V_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ norm $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ to get $\begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix}$

So $V = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} = V^T$

$U =$ eigvecs of AA^T : $AA^T = A^T A$ so eigvecs are same

$U = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$ $\Sigma = \begin{bmatrix} \sqrt{\lambda_1} & 0 \\ 0 & \sqrt{\lambda_2} \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$

2

a)

Given: $A = QBQ^T$ where $QQ^T = Q^T Q = I$

Let λ be a singular value. Then $\exists x \neq 0$ st $A^T A x = \lambda x$

$$\implies (QBQ^T)^T (QBQ^T) x = \lambda x$$

$$\implies (QB^T Q^T) (QBQ^T) x = \lambda x$$

$$\implies QB^T BQ^T x = \lambda x$$

$$\implies B^T BQ^T x = Q^T \lambda x$$

$$\implies B^T BQ^T x = \lambda Q^T x$$

Since Q is orthogonal, we know that Q and Q^T are nonzero and since x is nonzero, we know that $Q^T x \neq 0$ so let $y = Q^T x$ where $y \neq 0$

So $B^T B y = \lambda y$ therefore, λ is a singular value for B

b)

Converse: If they have the same singular values, then they are orthogonally similar

Let $A = [4]$ and $B = [-4]$. They have the same singular values, 4. For A and B to be orthogonally similar, we need Q to be positive and Q^T to be negative (or vice versa). However if Q is positive then its transpose is also positive and if Q is negative then its transpose is also negative. So it is not possible for A and B to be orthogonally similar

3

In SVD, the choice of the singular values being positive is conventional

A is nonsingular so we know that it is invertible and has rank m

Let λ be an eigen value for A. Since A is nonsingular, $\lambda \neq 0$

This means that $Ax = \lambda x$ so multiplying by A^{-1} to both sides,

$x = A^{-1}\lambda x \implies A^{-1}x = \frac{1}{\lambda}x$ so we know that the eigenvalues of A^{-1} are the reciprocal of the eigenvalues of A.

Since A is nonsingular, we know that $\sigma_1 \neq 0$.

By what we showed previously, we know that the singular values of A^{-1} are the reciprocal of the singular values of A

ie σ_1^{-1}

Since σ_1 was the smallest, σ_1^{-1} is the largest singular value for A^{-1} therefore

$$\sigma_1^{-1} = \frac{1}{\sigma_1}$$

A similar proof is that $\sigma_n^{-1} = \left(\frac{1}{\sigma_n}\right)^{-1} = \sigma_n$ and since Σ is a diagonal matrix, its inverse will be the reciprocal of its values. And the largest singular value of Σ^{-1} will be the reciprocal of the smallest singular value of Σ so $\sigma_n^{-1} = \sigma_n$

4

```
A=[1 2 3 4; 5 6 7 8; 9 10 11 12; 1 1 1 1; 3 2 1 0]
[U, S, V] = svd(A, 'econ');

rank = sum(diag(S) > 1e-5)
A_2_norm = max(S, [], 'all')

fprintf("The orthonormal basis of the range is the column vectors of the
following matrix")
range_A= U(:, 1:rank)

fprintf("The orthonormal basis of the null space is the column vectors of
the following matrix")
null_A = V(:, rank+1:end)
```

```
>> q4

A =

     1     2     3     4
     5     6     7     8
     9    10    11    12
     1     1     1     1
     3     2     1     0

rank =

     2

A_2_norm =

    25.6545

The orthonormal basis of the range is the column vectors of the following matrix
range_A =

    -0.2043    -0.5076
    -0.5137    -0.1885
    -0.8230     0.1306
    -0.0773     0.0798
    -0.1051     0.8267

The orthonormal basis of the null space is the column vectors of the following matrix
null_A =

    -0.3590     0.4137
     0.7870    -0.2840
    -0.4970    -0.6731
     0.0690     0.5434
```

5

For this, assume $Q = Q$ and $R = R$. I got lazy

$\implies$

Let $\text{rank} A = n$, then $\text{rank} A^T = n$ so $A^T A$ is nonsingular.

Then we know that $\det(A^T A) \neq 0$

We know that $\det(A^T A) = \det((QR)^T QR) = \det(R^T Q^T QR)$. Since Q is made up of orthonormal columns, $Q^T Q = I$ so $\det(R^T R) = \det(R^T) \det(R) \neq 0$. This is only possible if and only if $\det(R^T) \neq 0$ and $\det(R) \neq 0$.

So we now have $\det(R) \neq 0$, since R is a triangular matrix, we know that the determinant of R is the product of its main diagonal and since determinant of R is not 0, there must not be a 0 along the main diagonal. Proof complete

R has all nonzero entries on the main diagonal

This implies that R is nonsingular so $\text{rank}(R) = n$

By rank inequalities: $\text{rank}(A) = \text{rank}(QR) \geq \text{rank}(Q) + \text{rank}(R) - n$

Since Q is an orthogonal matrix, $\text{rank} Q = \min\{m, n\} = n$ and $\text{rank}(R) = n$ so

$\text{rank}(A) \geq n + n - n = n$

So now we have $\text{rank}(A) \geq n$. Since A is $m \times n$ matrix, we know its max rank is n thus $\text{rank}(A)=n$

6

Show that the columns of Q form an orthonormal basis for the Range of A .

So we know that the cols of Q are orthonormal by the defn of QR factorization and we know the cols of Q form the $\text{range}(Q)$ by base defn.

WTS $\text{range}(Q)=\text{range}(A)$. Will do this by containment and reverse containment

$$\text{range}(Q) \supseteq \text{range}(A)$$

Let $x \in \text{Range}(A)$ where $x \neq 0$, since $\text{rank} A=n \exists y \neq 0$ st $Ay=x$

$$\implies QRy = x, \text{ since } y \neq 0, Ry = z \neq 0 \text{ so } \forall x \in \text{Range}(A) \exists z \text{ st } Qz = x \text{ so}$$

$$x \in \text{range}(Q) \text{ so } \text{range}(A) \subseteq \text{range}(Q)$$

$$\text{range}(Q) \subseteq \text{range}(A)$$

Let $x \in \text{Range}(Q)$ where $x \neq 0$, then $\exists y \neq 0$ st $Qy=x$. Since R is upper triangular with nonzero entries along the main diagonal, it is nonsingular and therefore invertible. So we can write

$$Q = AR^{-1} \text{ so } Qy = AR^{-1}y = x \text{ so if we let } z = R^{-1}y, Az = x \text{ so then}$$

$$\forall x \in \text{Range}(Q), \exists z \text{ st } Az = x \text{ so } \text{range}(Q) \subseteq \text{range}(A)$$

$$\text{So } \text{range}(Q) \subseteq \text{range}(A) \subseteq \text{range}(Q)$$

$$\text{So } \text{range}(Q) = \text{range}(A)$$

7

Code:

```
function [Q, R] = clgs(A)

    n = size(A, 2);

    R = zeros(n);
    Q = zeros(size(A,1), n);
    for j = 1:n
        v = A(:, j);

        for i = 1:(j-1)
            R(i,j) = Q(:, i)' * A(:, j);
            v = v - R(i, j) * Q(:, i);
        end

        R(j,j) = norm(v);
```

```

        Q(:, j) = v / R(j,j);

    end
end

function [Q, R] = mgs(A)

    n = size(A, 2);

    R = zeros(n);
    Q = zeros(size(A,1), n);

    V = zeros(size(A));

    for i = 1:n
        V(:, i) = A(:, i);
    end

    for i = 1:n

        R(i, i) = norm(V(:, i));
        Q(:, i) = V(:, i) / R(i,i);

        for j = (i+1):n
            R(i, j) = Q(:,i)' * V(:, j);
            V(:, j) = V(:, j) - R(i, j) * Q(:, i);
        end
    end
end

epsillons = [1e-4, 1e-6, 1e-8];
format long

for e = epsillons
    A = [1, 1, 1; e, 0, 0; 0, e, 0; 0, 0, e];

    fprintf("Q and R for epsilon = %f using classical GSP", e)
    [Q, R] = clgs(A)
    fprintf("QR:")
    Q*R
    fprintf("Q^TQ:")
    Q' * Q

```

```
fprintf("Q and R for epsilon = %f using modified GSP", e)
[Q, R] = mgs(A)
fprintf("QR:")
Q * R
fprintf("Q^TQ:")
Q' * Q
end
```

```

../782/hw2 main* > matlab -batch q7
Q and R for epsilon = 0.000100 using classical GSP
Q =

    0.999999995000000    0.000070710677866    0.000040824828876
    0.000099999999500   -0.707106775883247   -0.408248287161463
           0           0.707106782954315   -0.408248290364194
           0           0           0.816496581608140

R =

    1.000000005000000    0.999999995000000    0.999999995000000
           0           0.000141421355884    0.000070710677866
           0           0           0.000122474487037

QR:
ans =

    1.000000000000000    1.000000000000000    1.000000000000000
    0.000100000000000   -0.000000000000000           0
           0           0.000100000000000           0
           0           0           0.000100000000000

Q^TQ:
ans =

    1.000000000000000    0.000000000000277    0.000000000000160
    0.000000000000277    1.000000000000000   -0.000000002264673
    0.000000000000160   -0.000000002264673    1.000000000000000

Q and R for epsilon = 0.000100 using modified GSP
Q =

    0.999999995000000    0.000070710677866    0.000040824829036
    0.000099999999500   -0.707106775883247   -0.408248288762828
           0           0.707106782954315   -0.408248288762829
           0           0           0.816496581608140

R =

    1.000000005000000    0.999999995000000    0.999999995000000
           0           0.000141421355884    0.000070710677588
           0           0           0.000122474487037

```

QR:
ans =

1.0000000000000000	1.0000000000000000	1.0000000000000000
0.0001000000000000	-0.0000000000000000	0
0	0.0001000000000000	0
0	0	0.0001000000000000

Q^TQ :
ans =

1.0000000000000000	0.0000000000000277	0.0000000000000160
0.0000000000000277	1.0000000000000000	-0.0000000000000000
0.0000000000000160	-0.0000000000000000	1.0000000000000000

Q and R for epsilon = 0.000001 using classical GSP
Q =

0.9999999999999500	0.000000707169643	0.000000408248287
0.0000010000000000	-0.707106781186017	-0.408211996414818
0	0.707106781186724	-0.408284583437063
0	0	0.816496579852289

R =

1.0000000000000500	0.9999999999999500	0.9999999999999500
0	0.000001414213562	0.000000707169643
0	0	0.000001224744873

QR:
ans =

1.0000000000000000	1.0000000000000000	1.0000000000000000
0.0000010000000000	0	0
0	0.0000010000000000	0
0	0	0.0000010000000000

Q^TQ :
ans =

1.0000000000000000	0.0000000000062862	0.000000000036290
0.0000000000062862	1.0000000000000000	-0.000051326775655
0.0000000000036290	-0.000051326775655	1.0000000000000000

Q and R for epsilon = 0.000001 using modified GSP

Q =

0.999999999999500	0.000000707169643	0.000000408284584
0.000001000000000	-0.707106781186017	-0.408248290463693
0	0.707106781186724	-0.408248290463693
0	0	0.816496580927794

R =

1.000000000000500	0.999999999999500	0.999999999999500
0	0.000001414213562	0.000000707106781
0	0	0.000001224744871

QR:

ans =

1.000000000000000	1.000000000000000	1.000000000000000
0.000001000000000	0	0
0	0.000001000000000	0
0	0	0.000001000000000

$Q^T Q$:

ans =

1.000000000000000	0.0000000000062862	0.0000000000036294
0.0000000000062862	1.000000000000000	-0.000000000000000
0.0000000000036294	-0.000000000000000	1.000000000000000

Q and R for epsilon = 0.000000 using classical GSP

Q =

1.000000000000000	0	0
0.000000010000000	-0.707106781186548	-0.707106781186548
0	0.707106781186548	0
0	0	0.707106781186548

R =

1.000000000000000	1.000000000000000	1.000000000000000
0	0.000000014142136	0
0	0	0.000000014142136

```

QR:
ans =

    1.0000000000000000    1.0000000000000000    1.0000000000000000
    0.0000000100000000                0                0
                0    0.0000000100000000                0
                0                0    0.0000000100000000

Q^TQ:
ans =

    1.0000000000000000   -0.000000007071068   -0.000000007071068
   -0.000000007071068    1.0000000000000000    0.5000000000000000
   -0.000000007071068    0.5000000000000000    1.0000000000000000

Q and R for epsilon = 0.000000 using modified GSP
Q =

    1.0000000000000000                0                0
    0.0000000100000000   -0.707106781186548   -0.408248290463863
                0    0.707106781186548   -0.408248290463863
                0                0    0.816496580927726

R =

    1.0000000000000000    1.0000000000000000    1.0000000000000000
                0    0.000000014142136    0.000000007071068
                0                0    0.000000012247449

QR:
ans =

    1.0000000000000000    1.0000000000000000    1.0000000000000000
    0.0000000100000000                0                0
                0    0.0000000100000000                0
                0                0    0.0000000100000000

Q^TQ:
ans =

    1.0000000000000000   -0.000000007071068   -0.000000004082483
   -0.000000007071068    1.0000000000000000   -0.0000000000000000
   -0.000000004082483   -0.0000000000000000    1.0000000000000000

~/782/hw2 main* > 

```

The modified GSP was significantly better. This is evident at the smaller epsilons where $Q^T Q$ is much close to I_n for the modified GSP compared to the regular